

LABORATORY OBSERVATION

Course	Full Stack Web Development
Course Code	23CM4121
Observation Task No.	1
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1. TITLE

Interactive Product Calculator Using HTML, CSS and JavaScript

2. AIM

To develop an interactive product calculator that accepts a product name, quantity and unit price and dynamically calculates the total price without refreshing the webpage.

3. OBJECTIVE

1. To create a product calculator interface using HTML.
2. To style the calculator using CSS.
3. To accept product name, quantity and price as input.
4. To calculate the total price using a JavaScript function.
5. To use event listeners for dynamic updates.
6. To validate user input and display appropriate error messages.
7. To update webpage content using DOM manipulation.

4. THEORY

JavaScript can interact with HTML through the Document Object Model (DOM). Input values are read from HTML elements, converted into numbers, validated, and used in a calculation. Event listeners allow the total to update whenever the user changes the product name, quantity or price, without reloading the page.

Application Flow: Enter Product Details → Validate Input → Calculate Quantity × Price → Update DOM → Display Total

5. ALGORITHM / PROCEDURE

1. Create an HTML document containing product name, quantity and price fields.
2. Create a Calculate Total button and an area for the order summary.
3. Use JavaScript to access the input and output elements.
4. Create calculateTotal() to read and validate the inputs.
5. Calculate total price as quantity × unit price.
6. Display the product name and total using DOM updates.
7. Attach click and input event listeners.
8. Open the webpage in a browser and test valid and invalid inputs.

6. PROGRAM

```
1: <!DOCTYPE html>
2: <html lang="en">
3: <head>
4: <meta charset="UTF-8">
5: <meta name="viewport" content="width=device-width, initial-scale=1.0">
6: <title>Interactive Product Calculator</title>
7:
8: <style>
9: * {
10: box-sizing: border-box;
11: }
12:
13: body {
14: font-family: Arial, sans-serif;
15: background-color: #f2f2f2;
16: margin: 0;
17: padding: 40px;
18: }
19:
20: .container {
21: width: 400px;
22: margin: auto;
23: padding: 25px;
24: background-color: white;
25: border-radius: 10px;
26: box-shadow: 0 0 10px rgba(0, 0, 0, 0.15);
27: }
28:
29: h1 {
30: text-align: center;
31: margin-bottom: 25px;
32: }
33:
34: .form-group {
35: margin-bottom: 18px;
36: }
37:
38: label {
39: display: block;
40: margin-bottom: 6px;
41: font-weight: bold;
42: }
43:
44: input {
45: width: 100%;
46: padding: 10px;
47: font-size: 16px;
48: border: 1px solid #aaa;
49: border-radius: 5px;
50: }
51:
52: button {
53: width: 100%;
54: padding: 11px;
55: border: none;
56: border-radius: 5px;
57: background-color: #333;
58: color: white;
```

6. PROGRAM (continued)

```
59: font-size: 16px;
60: cursor: pointer;
61: }
62:
63: button:hover {
64: background-color: #555;
65: }
66:
67: #errorMessage {
68: color: red;
69: margin-top: 15px;
70: }
71:
72: .result {
73: margin-top: 20px;
74: padding: 15px;
75: background-color: #f5f5f5;
76: border-radius: 5px;
77: }
78:
79: .result h2 {
80: margin-top: 0;
81: }
82:
83: .result p {
84: font-size: 17px;
85: }
86: </style>
87: </head>
88:
89: <body>
90: <div class="container">
91: <h1>Product Calculator</h1>
92:
93: <div class="form-group">
94: <label for="productName">Product Name:</label>
95: <input type="text" id="productName" placeholder="Enter product name">
96: </div>
97:
98: <div class="form-group">
99: <label for="quantity">Quantity:</label>
100: <input type="number" id="quantity" min="1" value="1">
101: </div>
102:
103: <div class="form-group">
104: <label for="price">Price per Unit (■):</label>
105: <input type="number" id="price" min="0" step="0.01" value="0">
106: </div>
107:
108: <button id="calculateButton">Calculate Total</button>
109:
110: <p id="errorMessage"></p>
111:
112: <div class="result">
113: <h2>Order Summary</h2>
114: <p>Product: <span id="displayProduct">-</span></p>
115: <p>Total Price: ■<span id="totalPrice">0.00</span></p>
116: </div>
```

6. PROGRAM (continued)

```
117: </div>
118:
119: <script>
120: // Get HTML elements from the DOM.
121: const productName = document.getElementById("productName");
122: const quantity = document.getElementById("quantity");
123: const price = document.getElementById("price");
124:
125: const calculateButton = document.getElementById("calculateButton");
126: const displayProduct = document.getElementById("displayProduct");
127: const totalPrice = document.getElementById("totalPrice");
128: const errorMessage = document.getElementById("errorMessage");
129:
130: // Calculate and display the total price.
131: function calculateTotal() {
132:   const name = productName.value.trim();
133:   const qty = Number(quantity.value);
134:   const unitPrice = Number(price.value);
135:
136:   // Validate the product name.
137:   if (name === "") {
138:     errorMessage.textContent = "Please enter a product name.";
139:     totalPrice.textContent = "0.00";
140:     displayProduct.textContent = "-";
141:     return;
142:   }
143:
144:   // Validate the quantity.
145:   if (qty <= 0 || isNaN(qty)) {
146:     errorMessage.textContent = "Quantity must be greater than 0.";
147:     totalPrice.textContent = "0.00";
148:     return;
149:   }
150:
151:   // Validate the price.
152:   if (unitPrice < 0 || isNaN(unitPrice)) {
153:     errorMessage.textContent = "Price cannot be negative.";
154:     totalPrice.textContent = "0.00";
155:     return;
156:   }
157:
158:   // Calculate total = quantity × price.
159:   const total = qty * unitPrice;
160:
161:   // Update the webpage without refreshing it.
162:   errorMessage.textContent = "";
163:   displayProduct.textContent = name;
164:   totalPrice.textContent = total.toFixed(2);
165: }
166:
167: // Run calculation when the button is clicked.
168: calculateButton.addEventListener("click", calculateTotal);
169:
170: // Update the total dynamically as inputs change.
171: quantity.addEventListener("input", calculateTotal);
172: price.addEventListener("input", calculateTotal);
173: productName.addEventListener("input", calculateTotal);
174: </script>
```

6. PROGRAM (continued)

```
175: </body>
176: </html>
```

7. CODE EXPLANATION

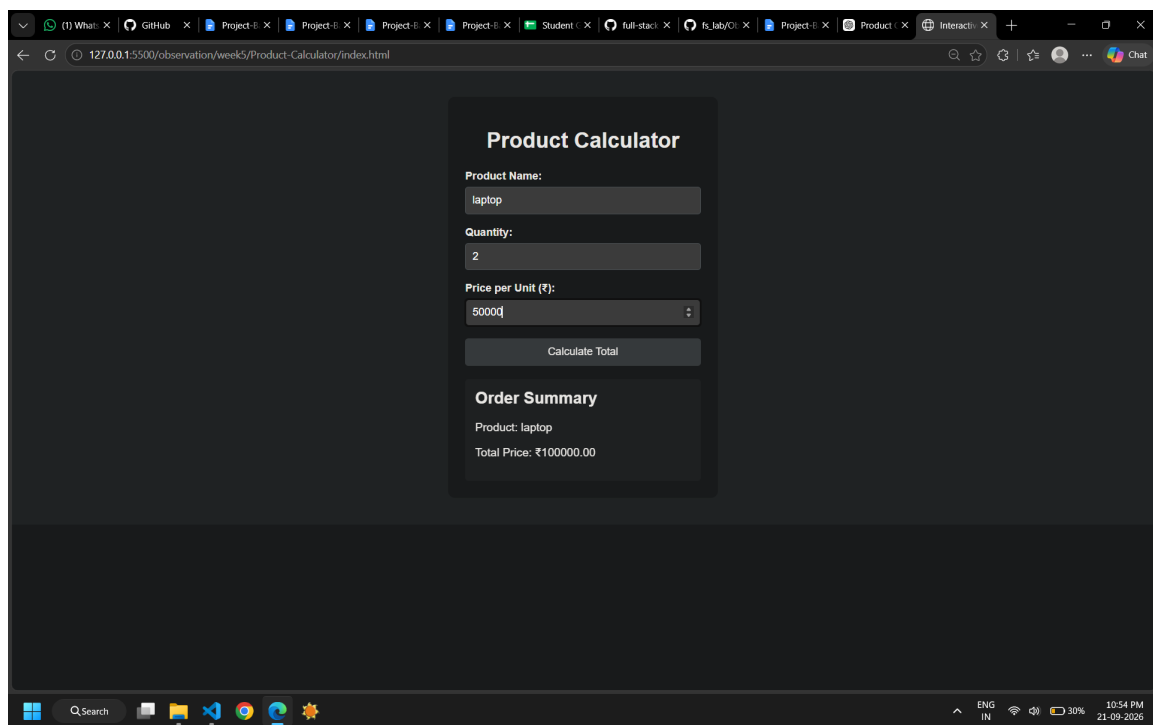
Code / Syntax	Explanation
getElementById()	Selects an HTML element using its unique ID.
Number()	Converts input values from strings to numbers.
trim()	Removes unnecessary spaces from the product name.
addEventListener()	Runs a function when a specified event occurs.
textContent	Changes the text displayed inside an HTML element.
toFixed(2)	Displays the calculated amount with two decimal places.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Laptop, quantity 2, price 50000	Total ₹100000.00	₹100000.00	Pass
TC-02	Change quantity to 3	Total updates to ₹150000.00	₹150000.00	Pass
TC-03	Empty product name	Error message displayed	Error displayed	Pass
TC-04	Quantity 0	Error message displayed	Error displayed	Pass

9. OUTPUT

Product Calculator output showing laptop, quantity 2, unit price ₹50000 and total price ₹100000.00.



Product Calculator

Product Name:

laptop

Quantity:

2

Price per Unit (₹):

50000

Calculate Total

Order Summary

Product: laptop

Total Price: ₹100000.00

10. RESULT

Thus, the Interactive Product Calculator was successfully developed using HTML, CSS and JavaScript. The application accepts product details, validates the input, calculates the total dynamically, and updates the result without refreshing the webpage.